Supplementary Material: Borrowing Strength Across Classes

Anonymous CVPR submission

1. Proofs

Throughout, $Q_\lambda = \alpha I + \lambda L_T$, $N_z = \mathrm{diag}(n)$, and for a fixed feature dimension j we write $d = d_{:j}$, $e = e_{:j}$ and $s = s_j$.

Derivation of ??. Given z, the terms of **??** that depend on D are $\sum_{i,k} z_{ik} \frac{(x_{ij} - t_{kj} - d_{kj})^2}{2\kappa\sigma_j^2}$ summed over j, plus $\frac{1}{2\kappa\bar{\sigma}^2} \sum_j d_{:j}^\top Q_\lambda d_{:j}$. Expanding the square around $\bar{x}_{kj}$ and dropping terms that do not depend on d gives, per dimension, $\frac{1}{2\kappa\sigma_j^2} \sum_k n_k (d_k - e_k)^2 + \frac{1}{2\kappa\bar{\sigma}^2} d^\top Q_\lambda d$. Multiplying by $\kappa\sigma_j^2$ and setting the gradient to zero yields $(N_z + s_j Q_\lambda) d = N_z e$, which is **??**. The matrix is positive definite whenever $\alpha > 0$ or $n_k > 0$ for some k in every connected component, so the solution is unique.

Proof of ??. With $\lambda = 0$ the system is diagonal: $(n_k + s\alpha) d_k = n_k e_k$. With $\alpha = 0$ and L_T the normalised Laplacian of the complete graph, $L_T = \frac{K}{K-1}(I - \frac{1}{K}\mathbf{1}\mathbf{1}^\top)$, whose null space is spanned by $\mathbf{1}$. As $\lambda \to \infty$ the minimiser of $\frac{1}{2}\sum_k n_k (d_k - e_k)^2 + \frac{\lambda s}{2} d^\top L_T d$ is forced into the null space, $d = \delta\mathbf{1}$, and minimising over δ gives $\delta = \sum_k n_k e_k / \sum_k n_k$. □

Proof of ??. Partition the system by A and P. Rows in A have $n_k = 0$ and $e_k = 0$, so they read $s\,(Q_{\lambda,AA} d_A + Q_{\lambda,AP} d_P) = 0$, which gives the harmonic extension $d_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP} d_P$ ($Q_{\lambda,AA}$ is positive definite when $\alpha > 0$, or when every component of A touches P). Substituting into the P rows gives $(N_{z,P} + s(Q_{\lambda,PP} - Q_{\lambda,PA} Q_{\lambda,AA}^{-1} Q_{\lambda,AP})) d_P = N_{z,P} e_P$, which is **??** on P with the Schur complement $Q_\lambda / Q_{\lambda,AA}$, the Kron reduction of the graph onto P [1]. □

Proof of ??. Given z, the per-dimension posterior of d is Gaussian with precision $(N_z + sQ_\lambda)/(\kappa\sigma_j^2)$, so its covariance is $\kappa\sigma_j^2 (N_z + sQ_\lambda)^{-1}$. For symmetric positive definite A and positive semi-definite B, $(A+B)^{-1} \preceq A^{-1}$, hence the diagonal entries satisfy $[(N_z + s\alpha I + s\lambda L_T)^{-1}]_{kk} \le [(N_z + s\alpha I)^{-1}]_{kk} = 1/(n_k + s\alpha)$. Strictness: $[(A + B)^{-1}]_{kk} = [A^{-1}]_{kk}$ for all k in a set C would require B to vanish on the corresponding block of the Woodbury expansion, which fails for any class k with $(L_T)_{kk} > 0$, that is, any class with at least one neighbour. □

Proof of ??. With $\lambda_L = 0$ the objective is, in z, a sum of N strictly convex functions on the simplex whose minimisers are the softmax in **??** (standard mean-field argument), in π a Dirichlet-regularised multinomial likelihood whose minimiser is **??**, in D a strictly convex quadratic (above) and in σ^2 a separable problem whose stationary point is the stated variance. Each block update therefore does not increase J, which is bounded below, so J converges. With $\lambda_L > 0$ the term $-\frac{\lambda_L}{2m_s} z^\top W_s z$ is handled by the linear bound $-z^\top W_s z \le -2 z^\top W_s z^{(t)} + z^{(t)\top} W_s z^{(t)}$, valid when W_s is positive semi-definite and used as a majoriser otherwise, as in [3, 5]. □

2. Online algorithm

Algorithm 1 GRAFT, online inference on a stream

Require: anchors T, graph L_T, forgetting $\varrho \in (0, 1]$; state $(n, S, q, N) \leftarrow 0$, $D \leftarrow 0$, $\pi \leftarrow \mathbf{1}/K$

1: **for** each batch $X^{(b)}$ **do**
2: $\quad \hat{y} \leftarrow \mathrm{softmax}(\tau X^{(b)} T^\top)$; E-step with current (D, π, σ^2) **??**
3: $\quad (n, S, q, N) \leftarrow \varrho\,(n, S, q, N) + \big(z^\top \mathbf{1},\ z^\top X^{(b)},\ \sum_i x_i^{\odot 2},\ |X^{(b)}|\big)$
4: $\quad$ M-steps of **??** from (n, S, q, N); re-run the E-step on $X^{(b)}$ (two rounds)
5: $\quad$ **return** posterior z of $X^{(b)}$
6: **end for**

3. Empirical Bayes details

With P the present classes, the marginal covariance of $e_{:j,P}$ in pseudo-count units is $\tilde{Q}_P^{-1} + N_{z,P}^{-1}$ where $\tilde{Q}_P$ is the Schur complement of Q_λ onto P, because marginalising a GMRF onto a subset of nodes replaces the precision by its Schur complement [2]. The evidence of **??** sums the d independent dimensions after whitening E by $\sqrt{\kappa\bar{\sigma}^2}$. One Cholesky factorisation of a $|P| \times |P|$ matrix per grid point suffices; the Schur complement requires one solve with $Q_{\lambda,AA}$, which is

sparse and well conditioned when $\alpha > 0$. The rule is disabled when $|P| < 3$, since the evidence is then nearly flat in λ.

4. Datasets and protocol details

Table 1. Benchmarks (CoOp test splits). K: vocabulary size; N_{test}: number of test images.

Dataset	K	N_{test}	Dataset	K	N_{test}
ImageNet	1000	50,000	Oxford Pets	37	3,669
SUN397	397	19,850	Flowers102	102	2,463
FGVC-Aircraft	100	3,333	Caltech101	100	2,465
EuroSAT	10	8,100	DTD	47	1,692
Stanford Cars	196	8,041	UCF101	101	3,783
Food101	101	30,300	ImageNet-V2 / S	1000	10,000 / 50,889
			ImageNet-A / R	200	7,500 / 30,000

The batch sampler draws K_{eff} uniformly in $[k_{\min}, k_{\max}]$, selects K_{eff} classes without replacement, pools their test images and draws B of them without replacement (all of them if fewer than B are available). The online sampler draws, for each class, Dirichlet(γ) proportions over K slots, assigns the class's shuffled images to slots accordingly, shuffles within slots and concatenates the slots; $\gamma = -1$ denotes the separate stream. These re-implementations follow the description of [4]; the released adapter also runs GRAFT inside the official harness so that the exact official samplers can be used.

5. Hyper-parameters

Symbol	Meaning	Value
τ	logit scale of the VLM	100
κ	likelihood temperature (inherited)	50
λ_y	exponent of the zero-shot prior (inherited)	1
λ_L, m_s	sample-graph weight and neighbours (inherited)	1, 3
α	anchor strength (pseudo-counts)	1
λ	graph coupling (pseudo-counts)	4
m, ρ	class-graph neighbours and bandwidth	5, 0.05
c	Dirichlet pseudo-count of π	0.5
$T_{\max}$, n_{inner}	outer and inner iterations	10, 3
n_{cg}	conjugate-gradient iterations	30
ϱ	online forgetting factor	1 (none)

6. Additional result tables

7. Per-dataset results

Per-dataset versions of the main tables are generated by the released scripts (`visualization/generate_tables.py`) and will be included here once the runs complete.

Table 2. **Backbones and distribution shifts**, low setting (B=64, $K_{\text{eff}} \in [2, 10]$), top-1 accuracy (%). IN: ImageNet; V2/S/A/R: its shifted variants.

Backbone	Method	IN	V2	S	A	R
CLIP ViT-B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
CLIP ViT-L/14	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
OpenCLIP B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD
SigLIP B/16	zero-shot	TBD	TBD	TBD	TBD	TBD
	StatA	TBD	TBD	TBD	TBD	TBD
	GRAFT	TBD	TBD	TBD	TBD	TBD

Table 3. **Trustworthiness** in the low setting (B=64), averaged over eleven datasets, CLIP ViT-B/16. ECE and NLL: lower is better; worst-decile accuracy: higher is better.

Method	ECE (%) ↓	NLL ↓	Worst-decile acc. (%) ↑
CLIP zero-shot	TBD	TBD	TBD
TransCLIP	TBD	TBD	TBD
StatA	TBD	TBD	TBD
MOON	TBD	TBD	TBD
Anchor-only control	TBD	TBD	TBD
GRAFT	TBD	TBD	TBD

Table 4. **Likelihood, optimisation and cost**, low setting. Time is the wall-clock per task of 64 images on one GPU with cached features (ImageNet, K=1000).

Variant	acc. (%)
default (shared diagonal Σ, no re-projection, c=0.5, κ=50, $T_{\max}$=10)	TBD
isotropic Σ	TBD
per-class diagonal Σ_k (as StatA)	TBD
re-project μ_k onto the sphere	TBD
$c \in \{0.1, 2\}$	TBD/ TBD
$\kappa \in \{25, 100\}$	TBD/ TBD
$T_{\max} \in \{3, 20\}$	TBD/ TBD
exact solver instead of conjugate gradients	TBD

Table 5. **Computational cost.** Asymptotic cost per outer iteration and measured time per task (B=64, ImageNet, one GPU).

Method	dominant cost	ms / task
zero-shot	$O(NKd)$	TBD
TransCLIP	$O(NKd + Nm_sK)$	TBD
StatA	$O(NKd + Nm_sK)$	TBD
MOON	$O(NKd + Nm_sK)$	TBD
GRAFT (CG)	$O(NKd + Nm_sK + n_{\text{cg}}Kd(1+m))$	TBD
GRAFT + EB	above $+ O(\lvert G\rvert K^3)$	TBD

References

[1] Florian Dörfler and Francesco Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Trans-*

actions on Circuits and Systems I, 60(1):150–163, 2013. 1

[2] Håvard Rue and Leonhard Held. *Gaussian Markov Random Fields: Theory and Applications*. Chapman and Hall/CRC, 2005. 1

[3] Maxime Zanella, Benoît Gérin, and Ismail Ben Ayed. Boosting vision-language models with transduction. In *NeurIPS*, 2024. 1

[4] Maxime Zanella, Clément Fuchs, Christophe De Vleeschouwer, and Ismail Ben Ayed. Realistic test-time adaptation of vision-language models. In *CVPR*, 2025. 2

[5] Imtiaz Ziko, Jose Dolz, Eric Granger, and Ismail Ben Ayed. Laplacian regularized few-shot learning. In *ICML*, 2020. 1